

A Train-Free Prospective Reframe for Sunk-Cost Decision Advice

Anonymous submission

Abstract

Decision-support language models should ignore unrecoverable costs, but prompt-only warnings can also distort choices toward abandonment when continuing has higher prospective value. We study this tension with a matched factorial benchmark drawn from BiasBuster-derived prompts, CANNOT/TUM escalation tasks, and InvestorBench-derived investment decisions, evaluating bare prompting, a generic warning, and a prospective-value reframe on completed hosted-backend matrices of 720 tasks. On the primary OpenAI Responses backend P1 matrix, the reframe asks the model to compare future benefits, future costs, outside options, and uncertainty before a forced continue-or-cut-loss answer, reaching 619/720 correct versus 598/720 for the generic warning and 574/720 for bare prompting. Task-paired analysis shows an advantage over bare prompting (delta 0.063, 95% CI [0.042, 0.083]) and over the generic warning (delta 0.029, 95% CI [0.009, 0.050]). A second gpt-5-mini matrix is weaker and not decisive against bare prompting, so the result supports a bounded benchmark and candidate decode-time check rather than broad endpoint-family robustness, mechanism, or deployment guarantee.

Introduction

Decision-support uses of large language models (LLMs) make cognitive-bias-like behavior more than a benchmark curiosity. A model that advises whether to continue a failing project, admit a candidate, or hold an investment should base its recommendation on prospective value, not on unrecoverable prior costs. LLM decision-support work now spans uncertainty-sensitive choices, sequential decision making, and contextual strategic decisions (Liu et al. 2024; Chen, Zhang, and Zhu 2024; Klissarov et al. 2024; Lorè and Heydari 2023). Human decision research has long treated sunk-cost and escalation-of-commitment effects as cases where prior investment distorts future choices (Staw 1976; Arkes and Blumer 1985). Prospect theory and status-quo bias provide related accounts of how losses, reference points, and defaults can shape decisions (Kahneman and Tversky 1979; Samuelson and Zeckhauser 1988). Recent LLM work has begun to ask whether language models show analogous cognitive effects in decision contexts (Shaki, Kraus, and Wooldridge 2023; Thorstad 2023; Koo et al. 2023; Echterhoff et al. 2024).

Sunk-Cost Decision Advice

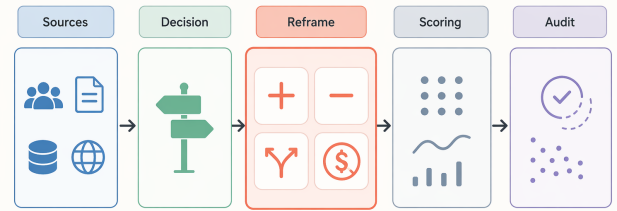


Figure 1: Contribution overview: a prospective-value reframe is evaluated as a train-free decision check on 720 matched tasks and 2160 scored method rows, reaching 619/720 correct on the primary P1 matrix while all remaining errors occur in near-tie rows.

Prompt-only mitigation work suggests that inference-time instructions can change model behavior without retraining, but generic reminders do not necessarily specify the decision basis a model should use (Ganguli et al. 2023; Xu et al. 2024; Anantaprayoon, Kaneko, and Okazaki 2025; Lyu et al. 2025). This paper studies a narrower question: when sunk-cost wording appears in decision advice, can a train-free prompt reframe reduce incorrect continue-or-cut-loss recommendations? We do not assume that LLMs uniformly imitate the human sunk-cost fallacy. The benchmark therefore treats both unnecessary continuation and unnecessary abandonment as failures against gold labels defined by prospective value.

We construct a matched factorial benchmark over three source families: BiasBuster-derived decision prompts, CANNOT/TUM escalation-of-commitment tasks, and InvestorBench-derived investment decisions. Each base item fixes future cost, future benefit, outside option, uncertainty, and the gold prospective decision, then varies neutral, sunk-cost, generic-negative, and accountability wording. The evaluated method is a prospective reframe: before the forced choice, the model is instructed to compare

only future benefits, future costs, outside options, and uncertainty, and to treat past costs as unrecoverable.

The completed evidence supports a focused result against the strongest prompt baseline on the primary run. On the primary OpenAI Responses backend P1 matrix, prospective reframe scores 619/720, generic warning scores 598/720, and bare prompting scores 574/720. The reframe is 0.063 higher than bare prompting with a 95% confidence interval of [0.042, 0.083], and 0.029 higher than the generic warning with a 95% confidence interval of [0.009, 0.050].

The contribution is intentionally bounded. We provide a behavioral diagnosis on a completed primary full matrix, a scoped second-backend matrix, a train-free decode-time intervention with a measured aggregate gain, and a failure analysis showing that near-tie cases remain difficult. We explicitly do not claim broad endpoint-family robustness, repeated-sample stability, benchmark-scale likelihood mechanisms, activation-level explanations, or broad source-context generality beyond the three executed source families.

Related Work

Sunk-cost and escalation-of-commitment research gives the normative target for this benchmark. Classic work on escalation shows how prior commitment can increase persistence (Staw 1976); sunk-cost experiments identify unrecoverable costs as a cause of irrational persistence (Arkes and Blumer 1985); prospect theory and status-quo bias frame related deviations around losses, reference points, and defaults (Kahneman and Tversky 1979; Samuelson and Zeckhauser 1988). Our benchmark uses these ideas to separate past-cost wording from prospective value.

LLM cognitive-bias work has moved from broad cataloging to decision-specific tests. Prior studies examine cognitive effects in LLMs (Shaki, Kraus, and Wooldridge 2023), benchmark LLMs as biased evaluators (Koo et al. 2023), and study cognitive bias in decision-making with LLMs (Echterhoff et al. 2024). Newer cognitive-bias benchmarks and domain studies broaden that line to multi-bias frameworks, medical settings, and instruction-tuned systems (Huang et al. 2026; Shaikh et al. 2024; Schmidgall et al. 2024; Saeedi, Goodarzi, and Canbaz 2024). Other recent work studies context framing (Robinson and Burden 2025), model personality and bias manifestation (He and Liu 2025), and human-bias amplification (Sun et al. 2025). We differ by isolating sunk-cost wording under fixed future value and treating over-abandonment as a first-class error.

Decision-domain studies show why source slices matter. Financial-bias and investment settings, conversational prediction of biased human choices, and behaviorally grounded framing studies all suggest that model responses can vary with domain and wording even when the underlying decision problem is similar (Hu and Zhao 2026; Pilli and Nallur 2026; Hwang et al. 2026). Evaluator-bias work similarly warns that LLM judgments can be inconsistent under surface changes (Stureborg, Alikaniotis, and Suhara 2024; Liu and He 2024). Our source-family matrix therefore treats BiasBuster-derived, CANNOT/TUM-

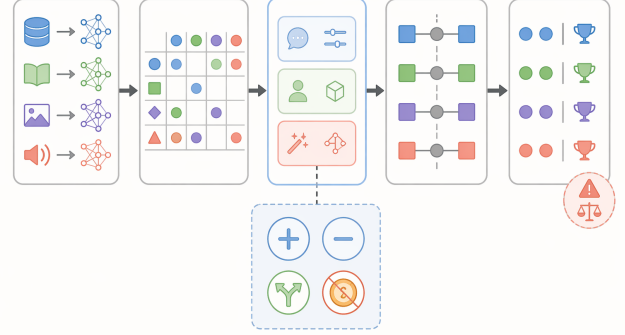


Figure 2: Evaluation pipeline: source tasks are converted into a factorial decision matrix, evaluated with bare prompting, a generic warning, and the prospective reframe, then summarized with paired deltas and a near-tie failure taxonomy.

derived, and InvestorBench-derived rows separately rather than reporting only an aggregate score.

Prompt-only mitigation is also relevant. Moral self-correction (Ganguli et al. 2023), perspective-taking prompts (Xu et al. 2024), intent-aware self-correction (Anantaprayoon, Kaneko, and Okazaki 2025), and self-adaptive cognitive debiasing (Lyu et al. 2025) all suggest that inference-time instructions can change biased outputs without training. Related self-debiasing, self-refinement, and prompt-tuning work provides additional context for train-free or low-training mitigation attempts (Feng et al. 2026; Mousavi et al. 2023; Tian et al. 2023). Our generic-warning baseline tests whether naming the issue is enough. The prospective reframe is more specific: it changes the represented decision basis rather than only asking the model not to be biased.

Mechanistic claims remain out of scope for this draft. Anchoring and mechanism-oriented studies (Huang et al. 2025; Takenami et al. 2025; Valencia-Clavijo 2025) motivate future white-box work, but the completed evidence here is behavioral forced-choice evidence only.

Benchmark and Prospective-Reframe Method

The benchmark contains 720 unique task ids, each evaluated under three method conditions, for 2160 scored rows. Source families are balanced: 240 tasks each from BiasBuster-derived, CANNOT/TUM-derived, and InvestorBench-derived decision settings. Future-value regimes are also balanced after method expansion: continue-positive, cut-loss-negative, and near-tie regimes each contribute 720 scored method rows. Wording variants include neutral, sunk-cost, generic-negative, and sunk-cost accountability conditions, and the sunk-cost dose has four levels.

The gold label for each item is defined by future value rather than past cost. A continue-positive item is correct when the model recommends continuing; a cut-loss-negative item is correct when the model recommends stopping; a

near-tie item is deliberately more fragile and is analyzed separately. This design avoids treating human-like escalation as the only possible failure. A model can be wrong by continuing a bad project or by abandoning a good one after receiving sunk-cost warnings.

The factorial controls are central to the paper’s evidence discipline. Neutral rows establish the decision without the sunk-cost cue. Sunk-cost rows add unrecoverable investment. Generic-negative rows separate sunk-cost specificity from broadly adverse wording. Accountability rows add responsibility pressure that could otherwise be confused with escalation of commitment. Dose levels vary the amount of unrecoverable investment, but the gold label remains tied to prospective value. These controls let the paper discuss decision-basis sensitivity without converting every observed difference into a sunk-cost claim.

We evaluate three methods on the same task ids. The bare prompt presents the decision item and asks for the forced choice. The generic-warning baseline tells the model not to let negativity, prior spending, ownership, or commitment pressure affect the answer and to use only stated future benefits, future costs, and the outside option. The proposed prospective reframe asks the model to compare net future value for continuing against cutting loss, treat all past costs as unrecoverable and irrelevant, and then answer with exactly one label.

The method is train-free. It changes only the decode-time instruction and does not update weights, train a reward model, or use reinforcement learning. The core mechanism is the explicit decision representation: future benefits, future costs, outside options, and uncertainty are foregrounded, while unrecoverable prior investment is excluded from the decision basis.

The method also intentionally avoids longer deliberation claims. It does not ask the model to write a chain-of-thought rationale, score alternatives with a learned verifier, or use an external calculator. The prompt-level change is narrow enough to compare directly with a generic warning under the same task ids and decoding settings. That makes the main comparison interpretable: when the reframe helps, the evidence supports changing the decision representation, not adding an arbitrary amount of extra text.

Experimental Setup

The primary evaluated system is OpenAI Responses backend P1, a hosted generation backend run with low reasoning effort, low text verbosity, and a 512-token output cap in a clean single-configuration matrix. We use P1 as the paper-facing identifier for this hosted backend because the service exposes the evaluated snapshot through the experiment manifest rather than a public model-card name. No temperature value is reported because the hosted interface for this backend did not expose temperature as a supported setting in the completed run. Responses were parsed as one of two natural forced-choice decisions, continue or cut loss, and scored against the prospective-value gold label. The primary completed matrix contains 2160 parsed method rows over 720 matched task ids. We also retain a completed gpt-5-mini

Setup item	Paper-facing configuration
Benchmark	720 matched task ids from three balanced source families
Methods	Bare prompt, generic warning, prospective reframe
Evaluated backend	OpenAI Responses backend P1; gpt-5-mini scoped second backend
Decoding budget	Responses API; low reasoning; low verbosity; 512-token cap for primary run
Sample policy	One parsed forced-choice response per method and task id
Metric	Accuracy against prospective-value gold labels; paired task-level deltas

Table 1: Experimental setup summary: the primary matrix evaluates three prompt conditions on the same 720 task ids under OpenAI Responses backend P1, with a scoped gpt-5-mini second-backend matrix retained for calibration.

matrix as scoped second-backend evidence; its reframe-vs-bare confidence interval crosses zero, so it is not used as a decisive robustness claim.

The primary metric is accuracy over forced choices. Because all three methods are run on the same task ids, comparative claims use paired task-level differences. We report paired mean accuracy deltas, normal-approximation 95% confidence intervals over paired score deltas, and exact two-sided sign tests over discordant task pairs.

The setup intentionally does not include repeated samples, alternate decoding, non-OpenAI-family endpoint runs, benchmark-scale likelihoods, activation analyses, or a full-scale reproduction of the original CANNOT/TUM paired metric. Calibration work on confidence and bias-sensitive outputs suggests that these omitted stability checks are not cosmetic; they are required before treating a prompt result as robust (Ghosh and Panday 2026; Sun et al. 2025). These omissions bound the claim to the completed forced-choice matrices rather than a broader robustness or mechanism result.

The run matrix is nevertheless complete for the bounded claim. Each of the 720 task ids appears under bare prompting, generic warning, and prospective reframe, so the paired comparisons do not compare different task mixtures. This matters for near-tie rows, where a small change in item mix could dominate the measured effect. The analysis therefore reports both aggregate accuracy and paired deltas over task ids rather than only unpaired method averages.

Results

Table 2 shows the central result and the paired comparisons. Prospective reframe is the highest-accuracy condition on the completed primary P1 matrix, with 619 correct choices and 101 failures. The reframe is 21 correct choices above the generic warning and 45 above bare prompting. In the paired block, prospective reframe is higher than bare prompting by 0.063, with 52 tasks where reframe is correct and bare prompting is wrong versus 7 tasks in the opposite direc-

Entry	Count	Score	Err./CI
Bare prompt	574/720	0.797	146 fails
Generic warning	598/720	0.831	122 fails
Prospective reframe	619/720	0.860	101 fails
Reframe vs bare	720	0.063	[0.042, 0.083]
Reframe vs warning	720	0.029	[0.009, 0.050]
Warning vs bare	720	0.033	[0.017, 0.049]

Table 2: Main result and paired deltas: prospective reframe reaches 619/720 correct on the primary P1 matrix and is higher than both bare prompting and the generic warning under task-paired intervals.

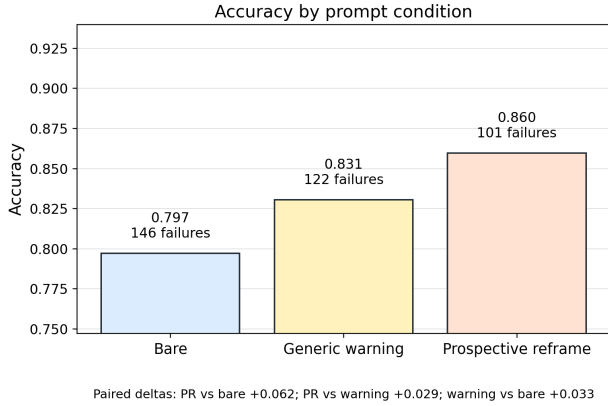


Figure 3: Paired-delta plot: prospective reframe is the highest-accuracy condition on the primary matrix, with positive task-paired deltas against both baselines.

tion. It is higher than the generic warning by 0.029, with 39 reframe-only correct tasks and 18 warning-only correct tasks. This supports the primary strongest-baseline claim for this backend, while the gpt-5-mini second-backend matrix is reported separately because its reframe-vs-bare interval crosses zero.

The family-level matrix shows that the aggregate result is positive across sources but uneven in size. Prospective reframe is highest on all three families, scoring 212/240 on BiasBuster-derived tasks, 211/240 on CANNOT/TUM-derived tasks, and 196/240 on InvestorBench-derived tasks. The largest family-level gain appears on CANNOT/TUM-derived rows; the InvestorBench slice has the smallest margin.

Table 3 is important because it prevents overclaiming from the overall average. The paper reports a complete three-family matrix and shows that the reframe is highest on each source family in the primary backend, while the magnitude varies by family. The strongest body claim is a task-paired strongest-baseline gain on the completed primary P1 matrix, not a universal model-family robustness claim.

Variant and Dose Slices

Appendix Table 6 separates the result by wording variant. The reframe is highest on all four variants in the primary run, with the largest margins on generic-negative and sunk-cost-accountability rows. This pattern is consistent with the aggregate story: explicitly representing future value helps most when the prompt contains adverse or accountability language.

The variant slices also make the failure story less sensational than an escalation-only narrative. Across all method failures, the taxonomy counts 100 generic-negative failures, 97 neutral failures, 88 sunk-cost failures, and 84 sunk-cost-accountability failures. Failures therefore do not concentrate only in the sunk-cost wording conditions. The decision remains difficult when generic negative language or neutral near-tie information is present, which supports the paper’s choice to discuss broader decision-basis sensitivity rather than only a classic sunk-cost effect.

Appendix Table 7 reports the generic-negative adjusted sensitivity analysis. Each cell is the paired continue-rate delta for a sunk-cost wording condition relative to generic-negative wording, using 180 paired rows per method and comparison. The aggregate deltas are negative: -0.056 and -0.050 for bare prompting, -0.072 and -0.061 for generic warning, and -0.050 and -0.022 for prospective reframe.

These deltas do not support a clean positive escalation effect. Instead, they support the narrower claim that wording can change decisions in fragile regimes, while the prospective reframe provides only a small aggregate accuracy gain under the measured forced-choice setup.

Dose slices lead to the same calibration. Failure counts by dose level are 94 for the zero-dose rows, 93 for low dose, 91 for medium dose, and 91 for high dose. If the completed matrix showed a monotone escalation process driven by past-cost magnitude, failures would be expected to grow clearly with dose. The observed failure counts do not support that stronger mechanism claim. They instead indicate that future-value regime and wording family matter more for the measured errors than a simple past-cost amount gradient.

Analysis and Failure Modes

The failure concentration is the most important caution. Near-tie cases account for all 369 failed scored rows in the primary run; continue-positive and cut-loss-negative rows are solved by all three methods under this backend. The result is therefore best read as a near-tie stress-test improvement rather than a claim that clear prospective-value rows remain difficult.

Near-tie failures still include both directions, but over-abandonment dominates. Across all primary-run failures, 333 rows have the gold decision continue and the parsed decision cut loss, while 36 have the gold decision cut loss and the parsed decision continue. The motivation is therefore not simply to prevent continuation: a sunk-cost warning can also push a model toward abandoning a prospectively positive option.

The generic warning baseline illustrates why the prospective reframe is useful but limited. Generic warning improves

Source family	Tasks	Backend/budget	Metric	Bare	Warning	Reframe
BiasBuster-derived	240	P1; low; 512 tokens	Accuracy	203/240	210/240	212/240
CANNOT/TUM-derived	240	P1; low; 512 tokens	Accuracy	184/240	199/240	211/240
InvestorBench-derived	240	P1; low; 512 tokens	Accuracy	187/240	189/240	196/240

Table 3: Cross-benchmark matrix: prospective reframe is highest on all three primary-run source families, with the largest margin on CANNOT/TUM-derived tasks.

aggregate accuracy over bare prompting in the primary P1 run, scoring 598/720 compared with 574/720. The prospective reframe reaches 619/720, adding 21 correct choices over the warning and 45 over bare prompting. Its advantage is meaningful for this matrix, but the remaining near-tie failures prevent treating it as a solved debiasing result.

The near-tie slice prevents overclaiming in a second way. Near-tie accuracy remains imperfect for all methods: 94/240 for bare prompting, 118/240 for generic warning, and 139/240 for prospective reframe. All methods solve the clear continue-positive and cut-loss-negative rows in the primary run, so the aggregate advantage comes entirely from the near-tie stress test. A deployment interpretation should therefore be cautious: this reframe improves ambiguous rows in the benchmark but is not a reliable solution for ambiguous real advice.

Two weak evidence sources are retained only as context. A CANNOT/TUM reproduction anchor covers five valid parsed control/treatment pairs and records mean escalation metric 0.1; this anchors protocol lineage but is not a full-scale baseline comparison. A 12-example open-model likelihood screen shows mean continue-label probability shifting from 0.710491 to 0.640424 under sunk-cost wording; this is a pilot signal, not benchmark-scale likelihood evidence.

The scope limits are binding for this paper. We completed a primary P1 matrix and a scoped gpt-5-mini matrix, but we did not complete repeated samples, alternate decoding, non-OpenAI-family endpoint runs, benchmark-scale label likelihoods, or activation analyses. The claims therefore apply to the completed forced-choice matrices over the three executed source families, not to broad model-family robustness.

The conceptual figures are intentionally low-text. They orient the reader to the contribution and evaluation flow, while the surrounding captions, tables, and prose carry the exact task counts, prompt conditions, and measured results.

Scope and Interpretation

The completed evidence supports a behavioral paper, not a mechanistic paper. The prospective reframe is not shown to activate a particular internal circuit or to suppress a known sunk-cost representation. The measured object is the final forced-choice recommendation under matched prompts. The comparison is therefore at the prompt-condition level: keeping task id, gold label, and supported decoding settings fixed, a future-value representation instruction is higher than both baselines on P1 and only scoped-supportive on the smaller gpt-5-mini run.

Table 4 explains why the result is useful but limited. The

Method	Clear	Tie	C→L	L→C
Bare	480/480	94/240	128	18
Warning	480/480	118/240	113	9
Reframe	480/480	139/240	92	9

Table 4: Regime and error-direction diagnostic: all methods solve clear-regime rows in the primary run, while prospective reframe improves near-tie correctness and reduces over-abandonment errors.

clear regimes are solved by all methods, while the hard near-tie rows separate the prompts: reframe reaches 139/240, compared with 118/240 for the warning and 94/240 for bare prompting.

The error-direction columns show why the benchmark is a decision-basis diagnostic rather than a one-direction fallacy detector. Across methods, many gold-continue rows are parsed as cut loss, while many gold cut-loss rows are parsed as continue.

The source-family slices constrain the conclusion. The benchmark draws from three curated prompt families rather than a deployment distribution. BiasBuster-derived items connect to cognitive-bias decisions, CANNOT/TUM-derived items anchor a published escalation protocol, and InvestorBench-derived items stress financial continuation and abandonment. The completed evidence does not show uniform source-family dominance, so the paper cannot claim cross-domain generalization.

Clear-regime and near-tie rows play different roles. Continue-positive and cut-loss-negative regimes are solved by all three methods, while near-tie rows drive the aggregate differences and remain mostly unsolved. That is a negative result for any claim that the reframe reliably resolves ambiguous sunk-cost advice.

The asymmetric failure direction is still informative. If all errors were wrong continuations, a generic instruction to ignore sunk costs might be enough. Instead, many failures resemble over-abandonment: the model is too willing to quit despite a gold continue label. This motivates keeping both continuation and abandonment errors in the benchmark.

The weak likelihood screen belongs here only as motivation. It used an open model and 12 examples, and it measured a mean decrease in continue-label probability under sunk-cost wording. That signal is consistent with over-abandonment, but it is not a benchmark-scale mechanism. The main paper therefore does not use likelihood values to explain the full matrix.

The CANNOT/TUM reproduction anchor is similarly

limited. It verifies that the project exercised the published source and metric structure, but its five parsed pairs are too small for a main baseline claim. The full matrix uses CANNOT/TUM-derived tasks as one of three balanced source families; it does not relabel the five-pair reproduction as decisive.

The generic warning baseline remains useful because it is a plausible prompt-only control. In the completed primary P1 matrix, it improves over bare prompting, but less than the prospective reframe. This matters: naming bias helps under this backend, but explicitly representing future value helps more. The paper therefore treats the reframe as a candidate check to test further, not as a validated guardrail.

The adjusted-sensitivity analysis is especially important for interpretation. In a classic sunk-cost framing, one might expect sunk-cost cues to increase continuation relative to a generic negative control. The completed aggregate slices do not show that pattern: paired continue-rate deltas for sunk-cost and accountability variants are negative relative to generic-negative wording. This does not mean sunk-cost wording is harmless. It means the measured behavior in this backend is bidirectional and context-sensitive.

The remaining failures are useful for the next experiment design. CANNOT/TUM-derived rows account for the largest family share of all failures, while the reframe’s strongest family-level advantage appears on BiasBuster-derived rows. Generic-negative and neutral variants account for many failures, not only explicitly sunk-cost variants. A future robustness run should therefore preserve neutral and generic-negative controls, keep near-tie rows explicit, and test whether the over-abandonment pattern persists under repeated samples and alternate model families. Those are prospective experiments, not claims made by this draft.

Finally, the completed result has a useful negative implication: a benchmark that only checks whether a model continues after seeing prior investment would miss much of the observed behavior. The gold-label construction here forces both sides of the decision to be possible. That design is why the paper can distinguish escalation from over-abandonment and why the final claim is a behavioral diagnosis of decision-basis sensitivity, not a one-direction fallacy detector.

Claim Calibration

The paper is organized around the claims supported by the completed forced-choice matrix rather than around broader hypotheses. Open-model likelihood margins, model-family comparisons, and possible activation contrasts remain useful future work, but the completed evidence does not support them as body claims. The paper therefore treats the completed forced-choice matrix as the evidence-bearing object and treats weaker auxiliary signals as scope notes.

Table 5 summarizes the resulting calibration. It helps prevent a common failure mode in papers about prompt interventions: a primary-backend result is easy to describe with language that sounds general. Here the numbers support a focused benchmark claim and a strongest-baseline prompt-comparison claim for the primary backend, plus scoped support from a weaker second backend, but not a general claim

Claim class	Draft treatment
Main method accuracy	Supported by completed scored rows and result tables
Paired method gains	Supported by task-paired significance table
Near-tie failures	Supported by failure taxonomy and failure rows
Original CANNOT/TUM baseline	Protocol anchor only, not a main-scale baseline
Likelihood shift	De-risk context only, not a benchmark-scale mechanism
Cross-model or sampling robustness	Omitted from main claims

Table 5: Claim calibration: only full-matrix forced-choice claims are promoted; weak or missing evidence classes are softened or omitted.

about all LLMs, all decision domains, or the internal causes of the behavior.

This calibration also affects how to read the generic-warning baseline. The generic warning is not a strawman; it encodes a plausible form of instruction-level debiasing that names the relevant failure mode. In this run, it improves over bare prompting, and the prospective reframe improves further. The scoped gpt-5-mini matrix, however, is not decisive against bare prompting, so the paper avoids claiming universal robustness.

That distinction matters: a completed run can still be insufficient for some claims. This paper uses the missing evidence classes to narrow the claim, not to discount the measured result. The strongest supported result remains the paired comparison over 720 task ids; the weak areas define the boundary around that result.

There is also a practical reason to keep the paper train-free. Many decision-support deployments use hosted systems where researchers cannot update model weights, inspect hidden states, or obtain calibrated log probabilities. A prompt-level check is attractive precisely because it can be applied at inference time. However, the same deployment constraint makes robust validation harder. Without repeated samples, model-family diversity, or backend-specific likelihoods, the paper cannot yet say how stable the reframe is under normal production variation.

The result should therefore be used as a diagnostic pattern: if a decision assistant is exposed to sunk-cost wording, a prospective-value self-check is a low-cost prompt to test, not a certified guardrail. In high-stakes use, it would need to be paired with domain-specific validation, uncertainty reporting, and human review. Those deployment controls are beyond the current run and are especially important because near-tie cases remain mostly unsolved.

The main quantitative pattern has three parts that should be interpreted together. Table 6 shows that the reframe is highest across the four wording variants in the primary matrix. Table 7 shows that sunk-cost wording does not create a positive aggregate continuation shift relative to the generic-negative control. Table 5 then marks the resulting bound-

ary: the paper supports a forced-choice benchmark and a small prompt-comparison result, not a one-direction sunk-cost replication or an internal-mechanism claim.

The row-level analysis also explains why the benchmark is useful despite the bounded intervention result. Continue-positive and cut-loss-negative rows test whether the method preserves clearer prospective decisions. Near-tie rows expose whether an intervention preserves the correct prospective comparison when the answer is not obvious from a single large payoff difference. The completed run shows that the reframe helps in the near-tie stress test but does not solve it.

The source-family slices sharpen that diagnostic reading. CANNOT/TUM-derived tasks show the largest reframe-vs-bare gain, while BiasBuster-derived and InvestorBench-derived tasks show smaller positive gains. This does not justify claiming that one family is intrinsically harder in general, because the families were instantiated through this benchmark construction and one backend. It does justify a measured analysis claim: under the completed construction, source-family slices materially affect the method comparison.

The variant analysis likewise changes what a follow-up should prioritize. The sunk-cost-accountability condition has fewer aggregate failures than the neutral and generic-negative conditions in the completed taxonomy. That is not the pattern a one-direction escalation story would predict. It suggests that some responsibility wording may make the decision basis clearer for this backend, or that the near-tie task templates interact with responsibility wording in a way that is not reducible to past-cost pressure alone. The paper does not choose between those explanations. It records the observed slice and uses it to narrow the claim to measured behavior.

The generic-negative control is especially important because it prevents a false positive. If a model changes its answer after reading sunk-cost wording, that change could reflect the unrecoverable-cost cue, but it could also reflect generic negative affect, risk language, or the presence of more adverse context. The adjusted continue-rate table compares sunk-cost and accountability variants against generic-negative wording rather than against only neutral rows. The aggregate adjusted deltas are not positive, so the paper cannot honestly say that sunk-cost cues increased continuation in the completed matrix. What it can say is that the prospective reframe improves correctness across variants while the aggregate continuation direction remains bidirectional.

This distinction also affects how the paper should be evaluated. A reader expecting a human-sunk-cost replication may see the negative adjusted continuation deltas as a failed replication. That would be a fair criticism of an escalation-only claim, but it is not a contradiction of the claim made here. The paper's claim is that sunk-cost-style decision advice requires checking the represented decision basis, because both over-continuation and over-abandonment can be wrong. The completed artifacts support that claim through the paired method gains, the near-tie failure concentration, and the failure-direction asymmetry.

The prompt intervention is consequently best viewed as

a candidate check rather than a debiasing proof. It asks the model to reconstruct the decision around future quantities and then produce a forced label. That is enough to produce a small gain against the generic warning in this matrix, but it is not enough to certify decisions whose relevant future quantities are missing, disputed, or value-laden. In such settings, the same prompt could faithfully compare incomplete information. This is why the experimental setup fixes future benefits, future costs, outside options, and uncertainty in the task text before scoring the model output.

The paper also avoids claiming that the prospective reframe is uniquely optimal. Other train-free prompts could express the same decision basis, and a tuned classifier or verifier could score decisions differently. The completed comparison is one concrete reframe against a bare prompt and one generic warning under matched task ids. The result supports the usefulness of explicit prospective-value representation, not the final wording of a prompt template. That wording should be treated as a reproducible intervention, not as a universally best instruction.

The practical takeaway is therefore conditional. If a decision prompt already states future benefits, future costs, outside options, and uncertainty, then a short prospective-value instruction is worth testing against simpler warnings. If those quantities are absent, disputed, or value-laden, the same instruction cannot recover missing decision information. The evidence in this paper is strongest as a benchmark-backed caution about prompt evaluation, not as a finished deployment recommendation.

Conclusion

This paper reports a bounded finding: on a completed 720-task primary P1 sunk-cost decision matrix, a train-free prospective reframe is higher than both a generic warning and bare prompting. The reframe reaches 619/720 correct, compared with 598/720 for the generic warning and 574/720 for bare prompting. The paired gain over bare prompting is 0.063 with a 95% confidence interval of [0.042, 0.083], and the paired gain over the warning is 0.029 with a 95% confidence interval of [0.009, 0.050]. The near-tie analysis is equally central: all 369 failures occur where future value is deliberately close, and a second gpt-5-mini matrix is not decisive against bare prompting, so the result should be read as a focused behavioral diagnosis and candidate decode-time check rather than a general robustness claim.

The negative results are part of the contribution. The adjusted continue-rate slices do not support a simple escalation story, and the auxiliary likelihood and CANNOT/TUM reproduction evidence is too small for a mechanism or full-baseline claim. Future work should test additional model families, repeated samples, likelihood-scale mechanisms, and white-box explanations before broadening the claim.

Limitations and Ethical Considerations

The main limitation is evidence scope. The primary full matrix uses one hosted backend, one supported Responses decoding configuration, and one parsed forced-choice sample per method and task id; the scoped gpt-5-mini matrix

adds a second OpenAI-family backend but not repeated sampling or endpoint-family diversity. That design is enough for the paired comparison reported in Tables 2 and 2, but it is not enough to estimate deployment stability under repeated sampling, model updates, alternative hosted systems, or open-weight model families. The paper therefore treats the prospective reframe as a candidate check to evaluate, not as a certified decision policy.

The benchmark labels are also stylized. They define correctness from prospective future value after fixing future benefit, future cost, outside option, and uncertainty in the task text. Real decision support often lacks those quantities, contains contested values, or involves stakeholders who disagree about the outside option. Prior work on candidate screening, finance, and clinical decision contexts shows why domain-specific validation matters before using LLM outputs in consequential decisions (Koh, Plata, and Chai 2023; Hu and Zhao 2026; Schmidgall et al. 2024). In such settings, a prompt that asks a model to compare future quantities may still faithfully compare incomplete or biased information. The ethical use of this benchmark is therefore diagnostic: it can reveal whether a model follows the stated prospective basis, but it should not be used to automate consequential continuation, hiring, medical, financial, or resource-allocation decisions without domain review.

The error pattern reinforces that caution. Near-tie rows remain mostly unsolved, and failures include both over-continuation and over-abandonment. A mitigation that only discourages continuation after sunk-cost wording could harm users when continuing is prospectively correct. Any applied system using this style of prompt should expose uncertainty, preserve human review for high-impact decisions, and validate on the local decision distribution rather than importing the aggregate numbers from this paper.

A second ethical limitation is that the benchmark is a controlled text evaluation rather than a study of affected users. The task construction is useful because it fixes the prospective quantities needed for scoring, but that same control removes institutional context, individual preferences, and downstream consequences. The results should therefore not be used to rank people, projects, investments, or treatments. They should be used to audit whether a model follows a stated future-value basis under simplified conditions.

The release boundary follows the same principle. The reproducibility target is to expose task metadata, prompt conditions, supported decoding settings, parser rules, and scored rows sufficient to recompute the tables. It should not expose private execution endpoints, local machine configuration, API credentials, or authoring workflow traces. This separation matters because reproducibility for a hosted-model benchmark requires enough information to check the scientific claim without turning local infrastructure details into reader-facing method content.

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Reproducibility Checklist

The reproduction target is to regenerate the reported tables and figures from the released scored rows and task metadata, then compile the anonymous AAAI source. The replay interface should expose the benchmark version, task count, prompt conditions, evaluated backend, supported Responses decoding settings, parser policy, metric definitions, and sample policy. Machine-specific execution settings are not part of the paper-facing method.

Appendix: Artifact Map

The supplementary package should contain scored rows, task metadata, aggregate result tables, paired-comparison summaries, failure-taxonomy summaries, and the figure assets used in the compiled paper. These artifact categories are sufficient to check the numerical claims without exposing machine-specific paths or authoring workflow metadata in the rendered manuscript.

Appendix: Supporting Slice Tables

Variant	Bare	Warn.	Refr.	Fail.
Neutral	141/180	148/180	154/180	97
Generic-neg.	138/180	150/180	152/180	100
Sunk cost	148/180	149/180	155/180	88
Sunk+acct.	147/180	151/180	158/180	84

Table 6: Variant slices: prospective reframe is highest on all four wording variants in the primary P1 matrix.

Method	Sunk cost	Account.	Pairs
Bare prompt	-0.056	-0.050	180
Generic warning	-0.072	-0.061	180
Prospective reframe	-0.050	-0.022	180

Table 7: Generic-negative adjusted continue-rate deltas: sunk-cost wording does not produce a positive aggregate continuation shift relative to generic-negative wording in the completed matrix.